



CANDIDATE – PLEASE NOTE!

PRINT your name on the line below and return this booklet with your answer sheet. Failure to do so may result in disqualification.

TEST CODE **01207010**

MAY/JUNE 2018

FORM TP 2018044

**CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

BIOLOGY

Paper 01 - General Proficiency

1 hour 15 minutes

05 JUNE 2018 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

Which of the following diseases is due to a dietary deficiency?

- (A) Malaria
- (B) Diabetes
- (C) Influenza
- (D) Anaemia

Sample Answer



The correct answer to this item is “Anaemia”, so (D) has been shaded.

5. If you want to change your answer, erase it completely before you fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.
7. Figures are not necessarily drawn to scale.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. Which of the following features is used to classify a group of organisms as Class Insecta?

(A) Size
(B) Shape
(C) Colour
(D) Number of segments

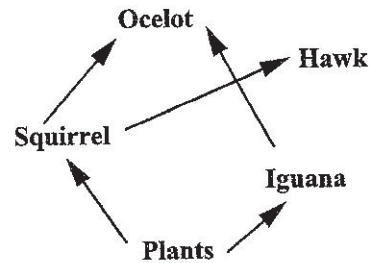
2. Which of the following sampling methods is MOST suitable for estimating the density of mushrooms in a garden?

(A) Quadrat
(B) Line transect
(C) Belt transect
(D) Mark/Release/Recapture

3. Living organisms, such as plants, are affected by abiotic factors which determine where they become established. Which of the following options lists some of these determining factors?

(A) Sediment size, shape and colour
(B) Sunlight availability, soil pH, minerals
(C) Parasitism, commensalism, mutualism
(D) Deforestation, slash and burn, shifting cultivation

Items 4–5 refer to the following food web.



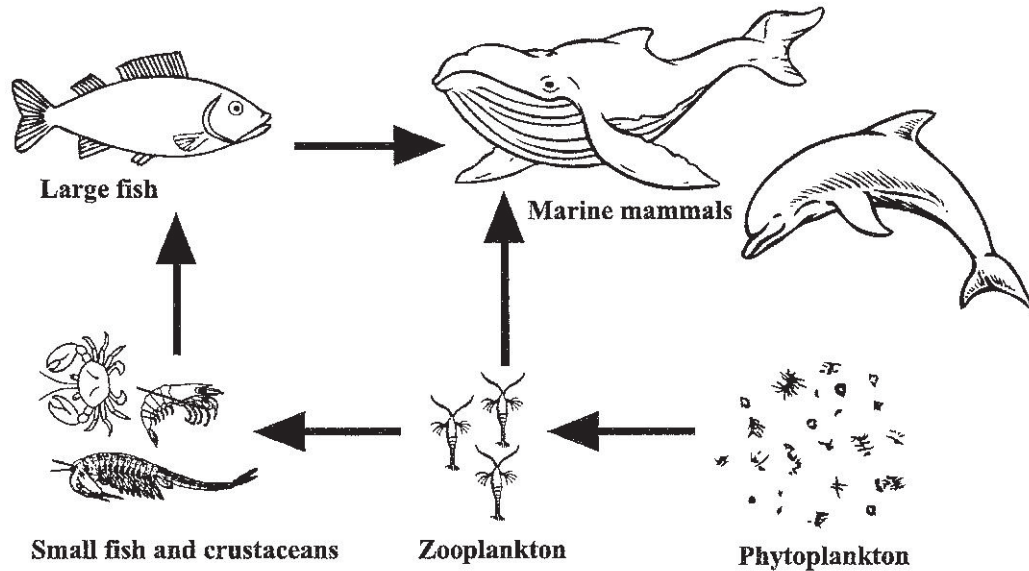
4. How many food chains are there in the food web shown above?

(A) 2
(B) 3
(C) 4
(D) 5

5. Which of the following statements describes the relationship where the ocelot eats an iguana and the iguana eats plants?

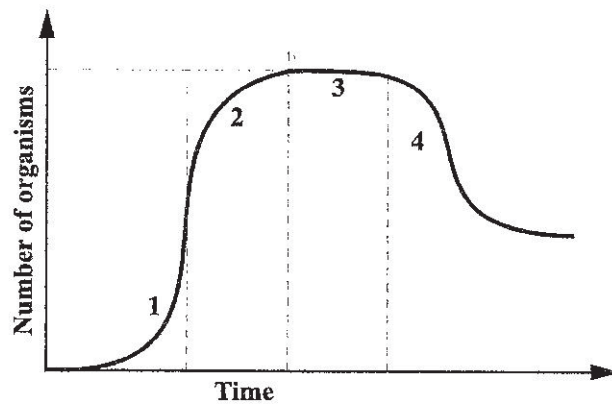
(A) Both the iguana and the ocelot are prey.
(B) Both the ocelot and the iguana are predators.
(C) The ocelot is the predator while the iguana is the prey.
(D) The ocelot is the prey while the iguana is the predator.

Item 6 refers to the following food web.



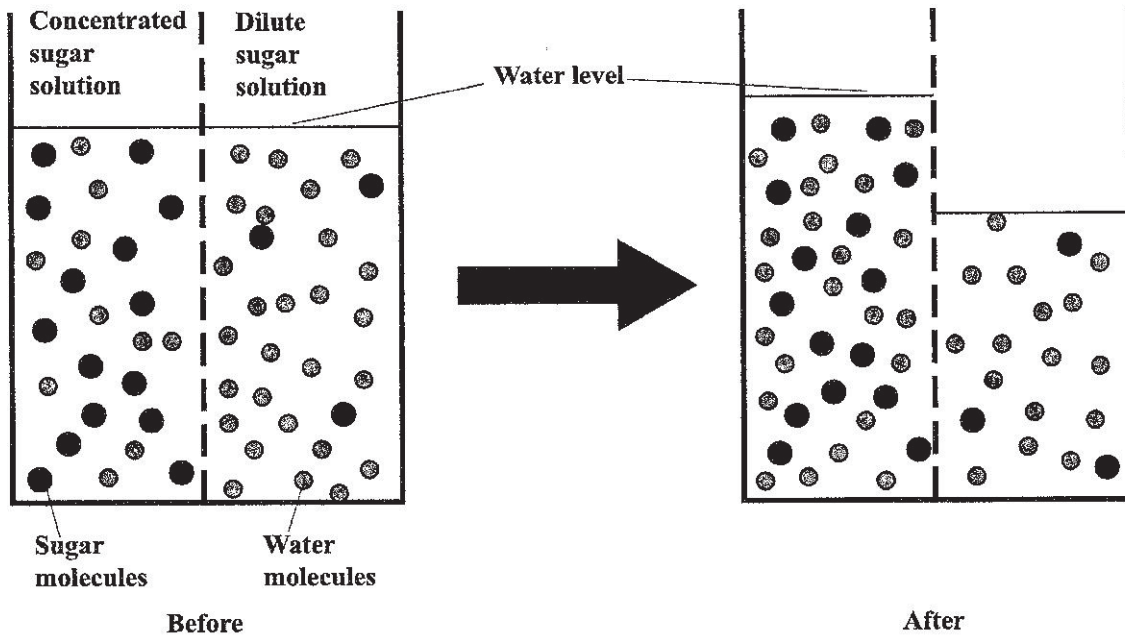
6. Trophic level 1 is represented by
- (A) phytoplankton
 - (B) zooplankton
 - (C) marine mammals
 - (D) small fish and crustaceans
-
7. Approximately 10% of the energy stored in food is available to the next organism in a food chain because
- (A) most of the energy is lost as heat during the process of respiration
 - (B) most of the energy is lost during the process of excretion
 - (C) there are fewer consumers than producers in a food chain
 - (D) many consumers compete for the same food source
8. Which of the following materials is considered to be non-biodegradable?
- (A) Grass cutting
 - (B) Sardine can
 - (C) Newspaper
 - (D) Cloth
9. Which of the following statements BEST explains why the coral reefs in the Caribbean are at risk?
- (A) There has been overfishing of the seas.
 - (B) The Caribbean is a popular tourist destination.
 - (C) There have been many hurricanes in the Caribbean.
 - (D) There has been an increase in the volume of sewage flowing into the reefs.
10. Replanting trees on a bare hillside is an example of
- (A) restoration and preservation
 - (B) preservation and conservation
 - (C) reafforestation and restoration
 - (D) conservation and afforestation

Item 11 refers to the following graph of population growth.



11. Phase 4 of the graph of population growth is MOST likely due to
- (A) disease resistance
 - (B) a high natural birth rate
 - (C) adequate food and space
 - (D) competition from invasive species
12. When compared to a cheek cell, a muscle cell contains more
- (A) vacuoles
 - (B) ribosomes
 - (C) mitochondria
 - (D) chromosomes

Item 13 refers to the following diagrams which show a process by which substances are moved into and out of cells.



13. Which of the following options correctly identifies and describes the process occurring above?

	Process	Description
(A)	Diffusion	Water moving from concentrated to dilute solution
(B)	Diffusion	Sugar moving from dilute to concentrated solution
(C)	Osmosis	Water moving from dilute to concentrated solution
(D)	Osmosis	Sugar moving from concentrated to dilute solution

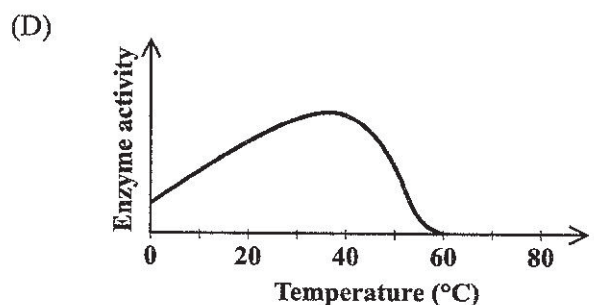
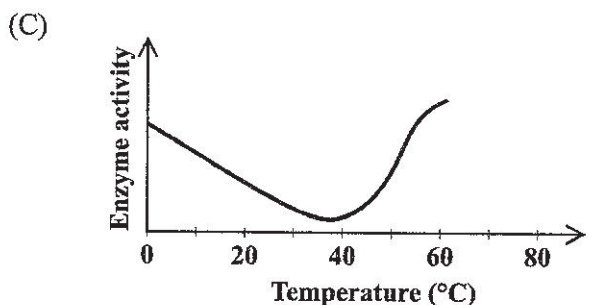
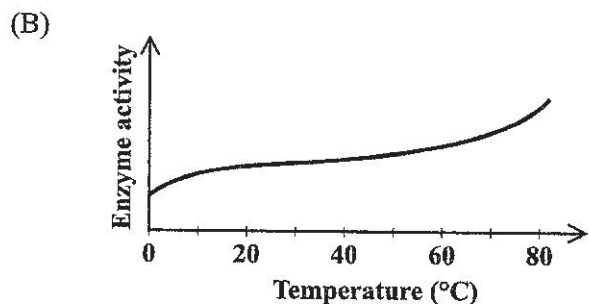
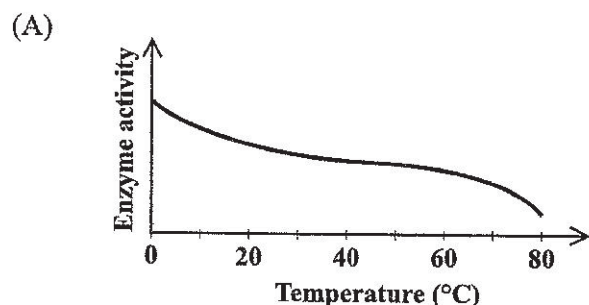
14. Which of the following functions is NOT correctly matched to the organelle?

	Function	Organelle
(A)	Osmotic control	Membrane
(B)	Protein synthesis	Chloroplast
(C)	Hereditary materials	Nucleus
(D)	Release of energy	Mitochondrion

15. A mature plant cell is different from a mature animal cell because the plant cell has

- (A) mitochondria
- (B) a cell membrane
- (C) a large permanent vacuole
- (D) a nucleus suspended in the cytoplasm

16. Which of the following graphs shows the effect of temperature on an enzyme-controlled reaction?



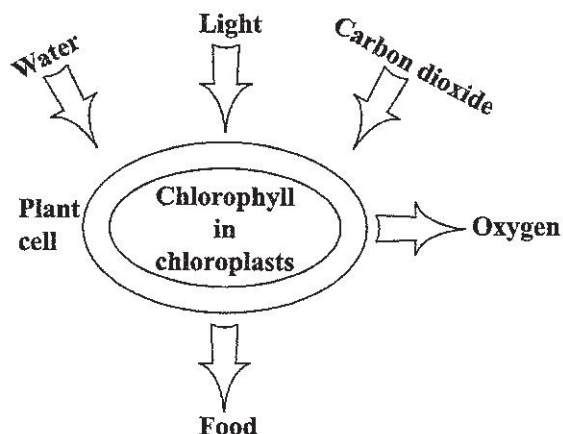
17. An enzyme is BEST defined as a catalyst which

- (A) increases the rate of a chemical reaction
- (B) decreases the rate of a chemical reaction
- (C) increases the rate of a chemical reaction but remains unchanged at the end of the reaction
- (D) decreases the rate of a chemical reaction but remains unchanged at the end of the reaction

18. Which of the following features does NOT enhance a leaf's ability to absorb solar energy?

- (A) Numerous chloroplasts in the palisade cells
- (B) Presence of a vascular bundle
- (C) Transparent cuticle
- (D) Large surface area

Item 19 refers to the following diagram which represents a metabolic process occurring in plants.



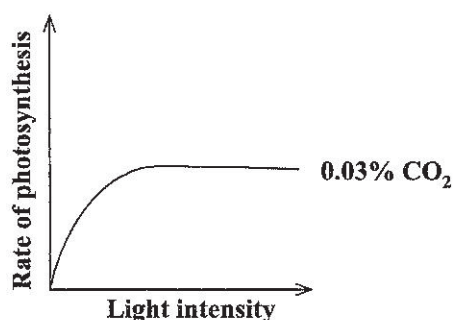
19. The food nutrient produced is

- (A) fat
- (B) glucose
- (C) protein
- (D) amino acid

20. Which of the following organisms is a saprophyte?

- (A) Cat
- (B) Amoeba
- (C) Mango tree
- (D) Bread mould

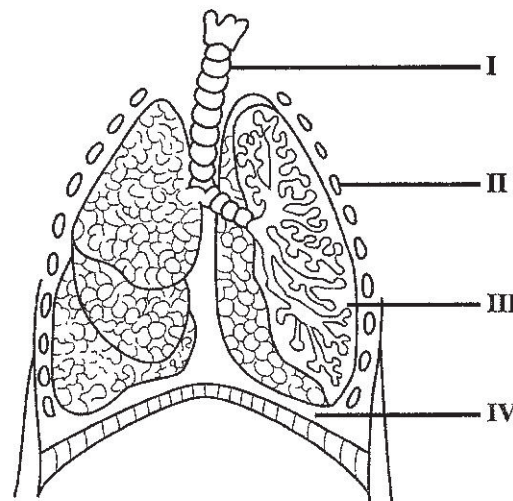
Item 21 refers to the following graph.



21. Which of the following statements BEST accounts for the shape of the graph of light intensity versus the rate of photosynthesis?

- (A) As the rate of photosynthesis increases, there is a decrease in carbon dioxide levels.
- (B) As light intensity increases, the rate of photosynthesis also increases, until a stationary phase exists, when denaturation of the enzymes occurs.
- (C) As light intensity increases, there is an increase in the rate of photosynthesis, until there is no further increase in the rate due to some other limiting factor.
- (D) As carbon dioxide levels gradually increase, there is a similar increase in the rate of photosynthesis, until a plateau phase exists where no further increase in carbon dioxide results in any further increase in photosynthesis.

Item 22 refers to the following diagram of the respiratory system showing structures in the respiratory system.



22. Which of the labelled parts represents a rib?

- (A) I
- (B) II
- (C) III
- (D) IV

23. Which of the following conditions is an effect of the nicotine found in cigarette smoke?

- (A) Increased heart rate
- (B) Increased beating of cilia
- (C) Reduced oxygen transport by the blood
- (D) Increased mucus production in the cells lining the respiratory passage

24. The steps listed below are involved in the clotting of blood.
- I. Fibrinogen is converted to fibrin.
 - II. Blood cells are trapped.
 - III. Platelets are activated.
 - IV. Blood vessels are damaged.
25. What is the MAIN storage product of plants?
- (A) Starch
 - (B) Sucrose
 - (C) Glucose
 - (D) Cellulose

Which of the following sequences is the correct order of the steps?

- (A) I → II → III → IV
 - (B) II → III → IV → I
 - (C) III → IV → I → II
 - (D) IV → III → I → II
-

26. Which of the following options correctly matches the gaseous exchange structures with the organism?

	Human	Fish	Plant
(A)	Trachea	Mouth	Stems
(B)	Alveoli	Gill filaments	Large leaves
(C)	Alveoli	Gill rakers	Young leaves
(D)	Trachea	Gill bars	Spongy mesophyll

27. Why is it difficult to develop a vaccine for the common cold?
- (A) Vaccines cannot be developed for viruses.
 - (B) The antigens on the common cold virus remain the same.
 - (C) The antigens on the common cold virus change frequently.
 - (D) Human beings cannot produce antibodies for the common cold.
28. Which of the following options BEST identify some of the transport substances in animals?
- I. Amino acids
 - II. Hormones
 - III. Sucrose
- (A) I and II only
 - (B) I and III only
 - (C) II and III only
 - (D) I, II and III

29. An amoeba obtains all the oxygen it needs by diffusion via its cell membrane, while a human needs to have special respiratory surfaces for this purpose. The BEST reason for this difference is that

(A) the amoeba does not require much oxygen
(B) oxygen cannot pass through the skin of a human
(C) a human requires a larger volume of oxygen
(D) a human's surface area to volume ratio is too small for diffusion to be effective

30. Translocation of sucrose in plants occurs via specialized vessels which

(A) are dead, narrow cells
(B) are abundant in mitochondria
(C) contain sieve pores in their end walls
(D) are thin, hollow tubes of lignin

31. Under which conditions will the rate of transpiration in plants be HIGHEST?

(A) Sunny and low wind speed
(B) Sunny and high wind speed
(C) Cloudy and low wind speed
(D) Cloudy and high wind speed

32. Persons who suffer from kidney disease are MOST likely to have

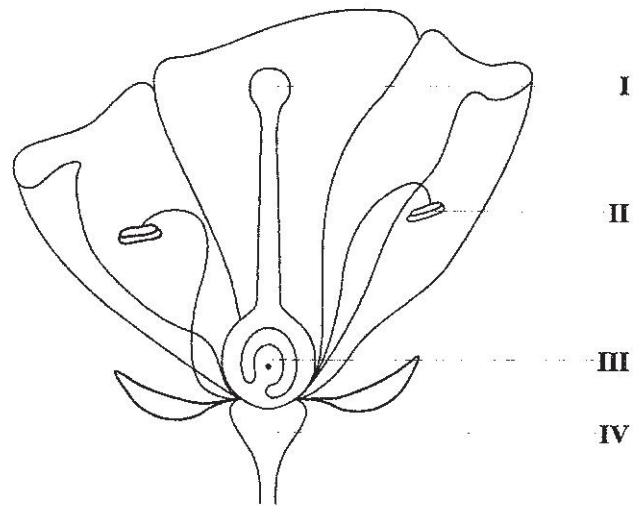
(A) a decreased concentration of glucose in the blood
(B) an increased concentration of glucose in the blood
(C) a decreased concentration of urea in the blood
(D) an increased concentration of urea in the blood

33. Locomotion is important to animals for

I. avoiding predators
II. acquiring food and shelter
III. reproduction

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

Item 34 refers to the following diagram of a flower.



34. Which of the labelled parts is responsible for the production of male gametes?

(A) I
(B) II
(C) III
(D) IV

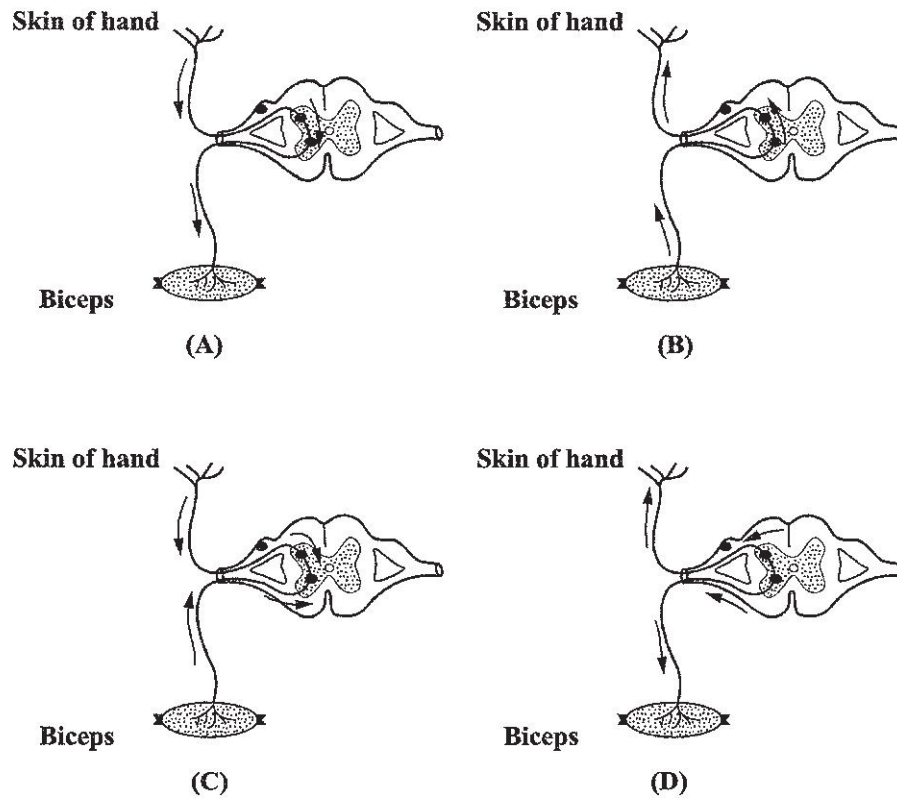
35. The BEST description for the role of excretion in living organisms is to

(A) produce urine
(B) produce faeces
(C) get rid of toxic waste
(D) get rid of excess water

36. The role of an effector is to

- (A) detect a stimulus
- (B) respond to a stimulus
- (C) create a nerve impulse
- (D) relay a nerve impulse

37. Which of the following diagrams shows the correct direction in which a nerve impulse passes through a reflex arc?



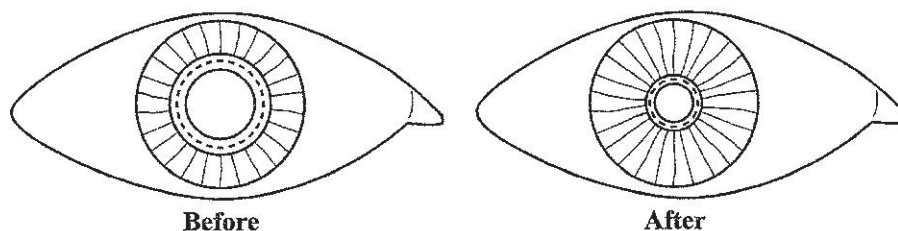
38. Which of the following phrases correctly describes tropic movement in plants?

- (A) Irreversible, whole or part movement
- (B) Reversible, whole or part movement
- (C) Irreversible, growth or part movement
- (D) Reversible, growth or part movement

39. Which of the following options BEST describes the terms 'stimulus' and 'response'?

	Stimulus	Response
(A)	A deliberate provocation of an organism	The movement of an organism
(B)	An organism's exposure to sunlight	The growth of the organism away from light
(C)	A wave of excitement in an organism	Locomotion of an organism
(D)	A detectable change in an organism's environment	An organism's reaction to a stimulus

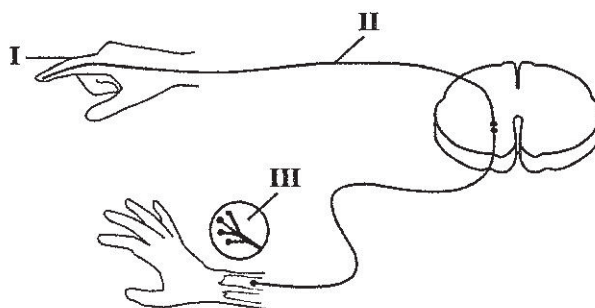
Item 40 refers to the following diagrams showing an eye's response to a stimulus.



40. The response is MOST likely brought about by

- (A) an increase in the light intensity
- (B) a decrease in the light intensity
- (C) an object being brought nearer
- (D) an object being moved further away

Item 41 refers to the following diagram.



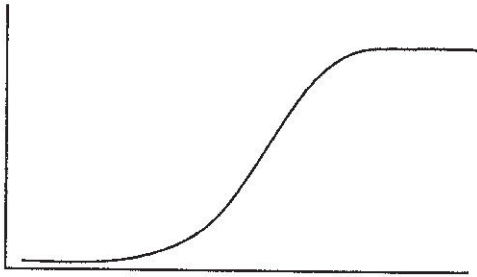
41. Which of the following options correctly identifies the structures labelled I, II and III?

	I	II	III
(A)	Receptor	Relay nerve	Effector
(B)	Skin	Intermediate nerve	Muscle
(C)	Sensory nerve	Spinal nerve	Effector
(D)	Receptor	Spinal nerve	Effector

42. The function of the choroid layer in the eye is to

- (A) focus most light rays
- (B) prevent internal reflection
- (C) maintain the shape of the eyeball
- (D) control the amount of light entering the eye

Item 43 refers to the following graph which illustrates a measurement of growth in living organisms.



43. Which of the following is an INCORRECT label for the y-axis?

- (A) Length
- (B) Mass
- (C) Number of leaves
- (D) Units of time

Item 44 refers to the following activities involved in the menstrual cycle.

- I. Repair of the uterine lining
- II. Ovulation
- III. Shedding of the uterine lining
- IV. Development of the Graafian follicle

44. Which of the following sequences is the correct order of activities?

- (A) I → II → III → IV
- (B) II → I → III → IV
- (C) IV → I → II → III
- (D) IV → III → II → I

45. The following statements describe the processes taking place within a seed during germination.

- I. Embryo uses food to develop radicle and plumule.
- II. Enzymes break down proteins into amino acids.
- III. Soluble products move into the embryo.

Which of the following sequences is the correct order of the processes?

- (A) I → II → III
- (B) I → III → II
- (C) II → I → III
- (D) II → III → I

46. The production of new organisms from one parent only is known as

- (A) sexual reproduction
- (B) asexual reproduction
- (C) mitosis
- (D) meiosis

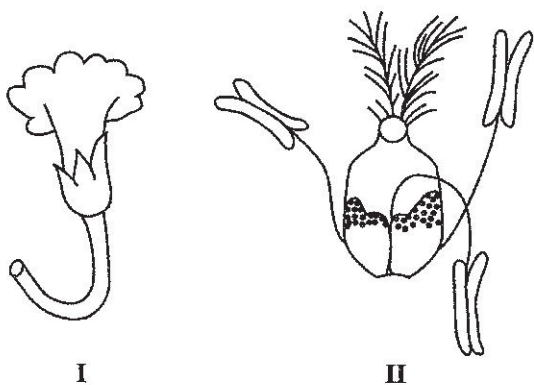
47. Which of the following forms of birth control is MOST likely to be 100% effective?

- (A) Condom
- (B) Diaphragm
- (C) Tubal ligation
- (D) Birth control pill

48. A vector is defined as an organism that

- (A) bites humans
- (B) causes diseases
- (C) lives on other organisms
- (D) transmits disease organisms

Item 49 refers to the following diagrams of flowers from two different types of plants.



49. Which of the following statements is true for the flowers?

- (A) I is pollinated by a humming bird; II is pollinated by the wind.
- (B) Both are pollinated by a humming bird.
- (C) Petals are absent from both I and II.
- (D) Both are wind pollinated.

50. A 60-year-old man has been diagnosed with hypertension as well as diabetes. Which of the following treatments would be MOST effective for the diseases?

- (A) Exercising and lowering the intake of salt and refined carbohydrates
- (B) Spending more time watching television to relax
- (C) Starting a rigorous exercise programme
- (D) Taking insulin injections

51. Which of the following BEST describes a chromosome?

- (A) Two forms of the same gene
- (B) Structure made up of DNA wrapped around histones
- (C) Nucleic acid that contains all genetic information
- (D) Part of the DNA which carries genetic information to produce a protein

52. As a result of mitosis each daughter cell has

- (A) a variable number of chromosomes
- (B) twice the number of chromosomes as the parent cell
- (C) the same number of chromosomes as the parent cell
- (D) half the number of chromosomes as the parent cell

53. Which of the following processes will result in genetically identical offspring?

- (A) Cloning
- (B) Mutation
- (C) Fertilization
- (D) Crossing over

54. A man with blood group A married a woman with blood group B. They had two children, EACH with blood group AB. The alleles A and B are described as being

- (A) homozygous
- (B) codominant
- (C) recessive
- (D) dominant

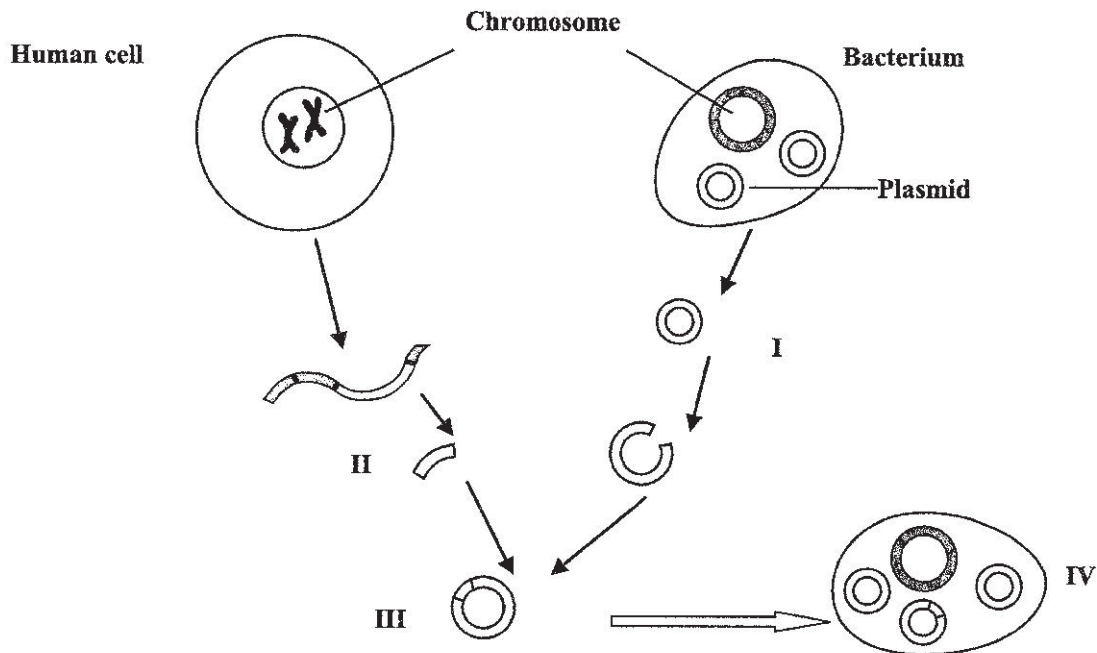
55. Variation in a population is due to
- I. mutation
 - II. crossing over
 - III. asexual reproduction
- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III
56. Which of the following features is NOT a characteristic of meiosis?
- (A) Maintenance of the diploid number
(B) Genetic recombination of alleles
(C) Crossing over of chromosomes
(D) Random assortment of genes
-
57. Tarzan and Jane, both of whom are heterozygous for blood groups A and B respectively, can have offspring that are likely to be
- (A) 100% AB
(B) 50% A: 50% AB
(C) 25% AB: 50% B: 25% O
(D) 25% A: 25% B: 25% AB: 25% O

58. Which of the following descriptions is true about natural and artificial selection?

	Natural Selection	Artificial Selection
(A)	Occurs in domestic populations	Occurs in natural populations
(B)	Involves genetic modification	Largely controlled by the environment
(C)	Produces great biological diversity	Produces very different organisms from natural populations
(D)	Is a faster process	Is a slower process

59. A species is BEST defined as a group of organisms that
- (A) can interbreed and produce fertile offspring
(B) can interbreed and produce many offspring
(C) are physically similar
(D) cannot interbreed

Item **60** refers to the following diagram which shows stages I, II, III and IV in the production of insulin by genetic engineering.



60. In which of the stages above can insulin be produced by the bacterium?

- (A) I
- (B) II
- (C) III
- (D) IV

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.